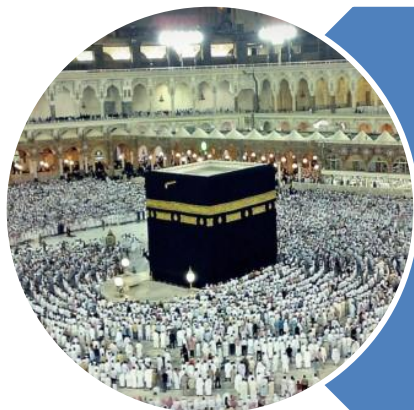


رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. **(Quran 20 : 25-28)**

Linked List

Variants or Types

Limitations of a Single linked list

- Insertion at the front is **$O(1)$**
- insertion at other positions is **$O(n)$**
- Insertion is convenient only **after a referenced node**
- Removing a node requires a reference to the **previous node**
- We can traverse the list **only in the forward** direction

C

P

C

S

2

0

4

4

Linked List Types

- **Single Linked List**
- Circular Linked List
- Double Linked List
- Circular Double Linked List

C

P

C

S

2

0

4

5

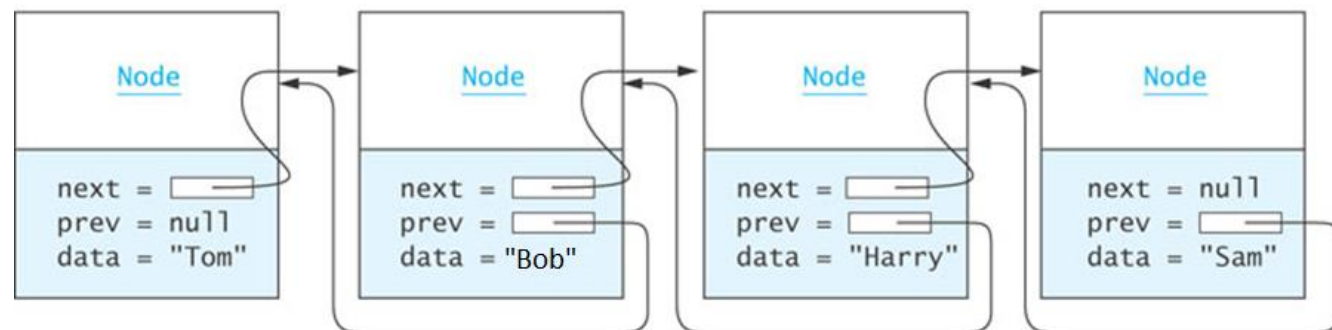
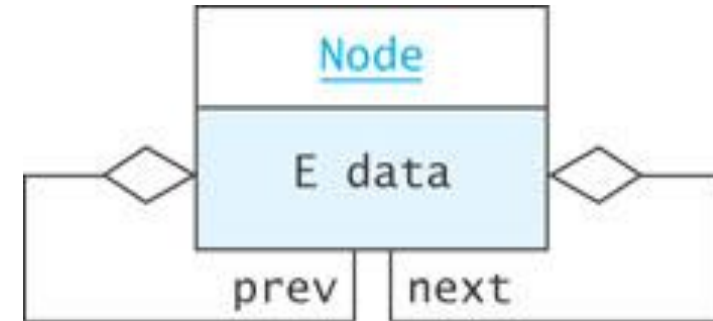
Linked Lists - Circular

- Circular Linked List
 - Last node **next** contains the address of **first** node
 - None of next part contains **null**
 - Visit all elements from **any starting point**
- **Disadvantage:** Must avoid an infinite loop



Linked Lists - Double

- Double Linked List
 - Two ways connectivity
 - **next** contains address of **successor** node
 - **prev** contains address of **predecessor** node
 - **next** of **last** node contains **null**
 - **prev** of **first** node contains **null**



C

P

C

S

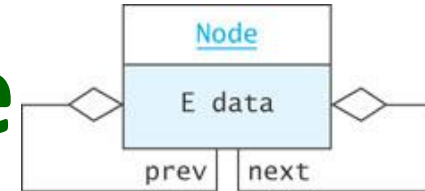
2

0

4

7

Double Linked List - Node

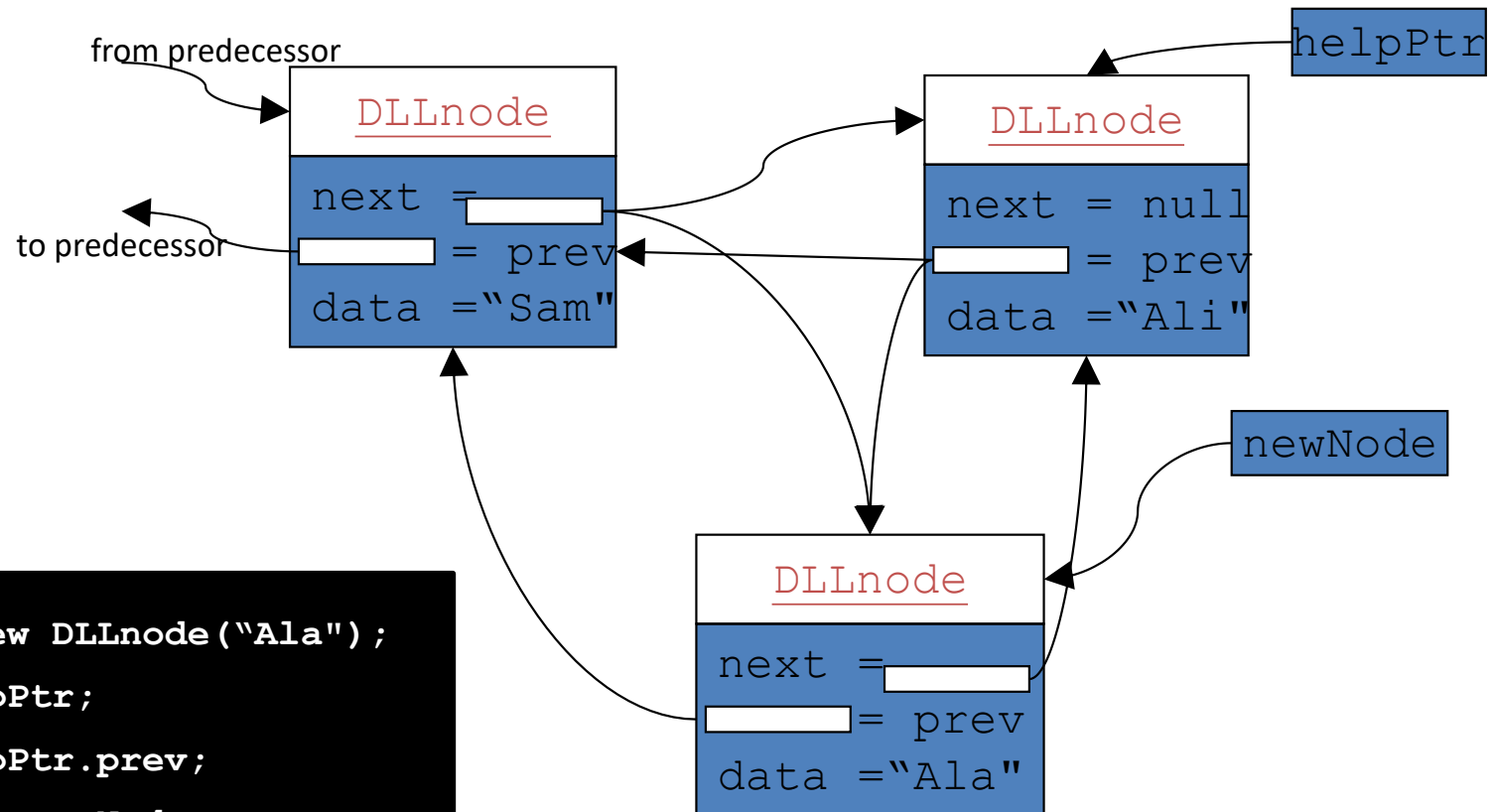


```
public class DLLnode {
    public int data;
    public DLLnode next;
    public DLLnode prev;

    public DLLnode(int i) {
        this(i, null, null);
    }

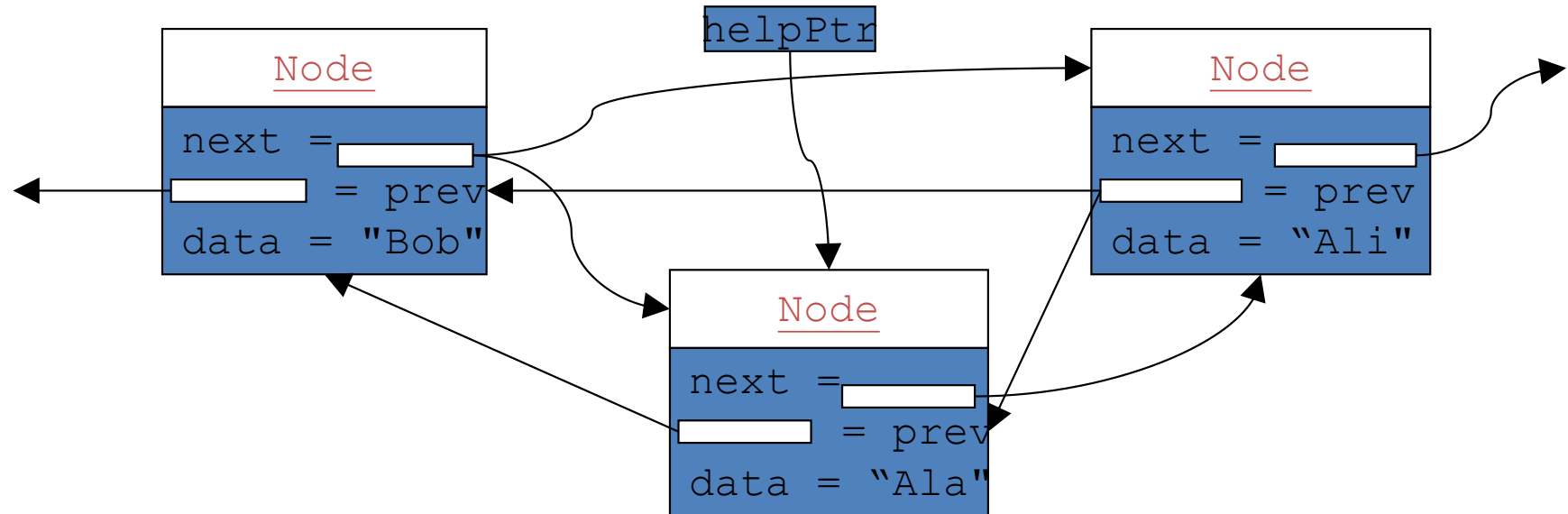
    public DLLnode(int i, DLLnode n, DLLnode p) {
        data = i;
        next = n;
        prev = p;
    }
}
```

Double Linked List - Insertion



```
DLLnode newNode= new DLLnode("Ala");
newNode.next = helpPtr;
newNode.prev = helpPtr.prev;
helpPtr.prev.next = newNode;
helpPtr.prev = newNode;
```

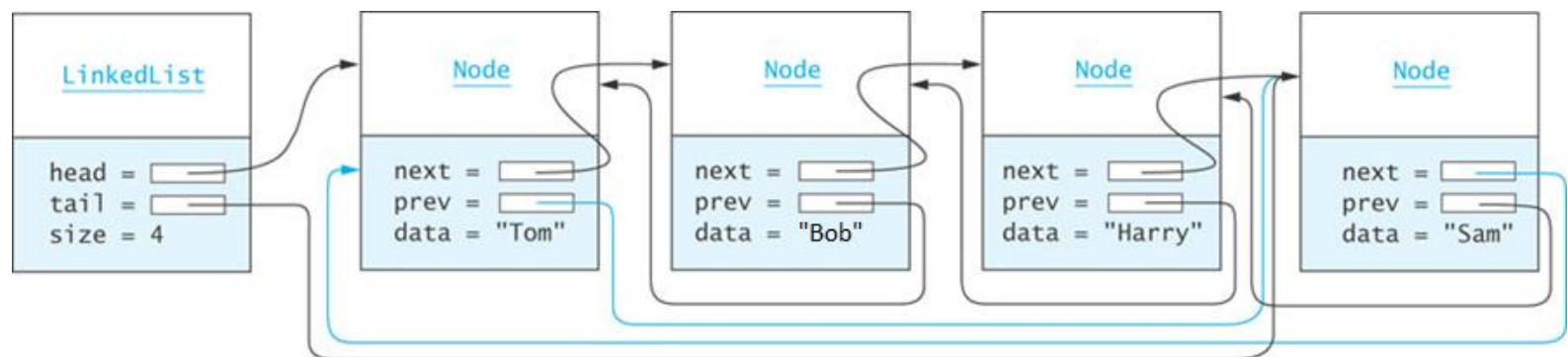

Double Linked List - Deletion



```
helpPtr.prev.next = helpPtr.next;
helpPtr.next.prev = helpPtr.prev;
```

Double Linked List - Circular

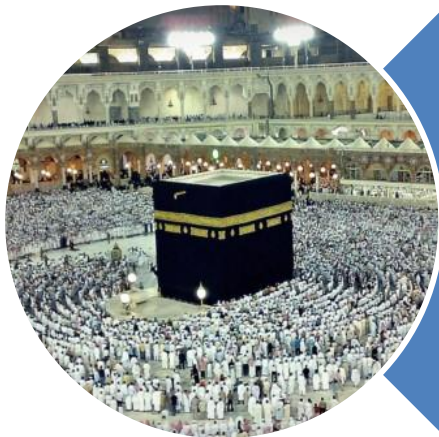
- **next** contains address of **successor** node
- **prev** contains address of **predecessor** node
- **next** of **last** node contains address of **first**
- **prev** of **first** node contains address of **last**



Double Linked List – Misc.

- A double-linked list object has data fields:
 - head (a reference to the first list Node)
 - tail (a reference to the last list Node)
- Insertion at either end is **$O(1)$**
- insertion elsewhere is still **$O(n)$**





Allah (Alone) is Sufficient for us,
and He is the Best Disposer of
affairs (for us). **(Quran 3 : 173)**